

EE263 Homework 2

Summer 2026

- 1. Halfspace.** Suppose $a, b \in \mathbb{R}^n$ are two given points. Show that the set of points in $\mathbb{R}^n$ that are closer to a than b is a halfspace, *i.e.*:

$$\{x \mid \|x - a\| \leq \|x - b\|\} = \{x \mid c^\top x \leq d\}$$

for appropriate $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$. Give c and d explicitly, and draw a picture showing a , b , c , and the halfspace.

- 2. Proof of Cauchy-Schwarz inequality.** You will prove the Cauchy-Schwarz inequality.

- a) Suppose $a \geq 0$, $c \geq 0$, and for all $\lambda \in \mathbb{R}$, $a + 2b\lambda + c\lambda^2 \geq 0$. Show that $|b| \leq \sqrt{ac}$.
- b) Given $v, w \in \mathbb{R}^n$ explain why $(v + \lambda w)^\top (v + \lambda w) \geq 0$ for all $\lambda \in \mathbb{R}$.
- c) Apply (a) to the quadratic resulting when the expression in (b) is expanded, to get the Cauchy-Schwarz inequality:

$$|v^\top w| \leq \sqrt{v^\top v} \sqrt{w^\top w}.$$

- d) When does equality hold?

- 3. (Optional) Norm inequalities.** Show that $\|a + b\| \geq \|a\| - \|b\|$.

- 4. (Optional) Input-output angle relationships.** Suppose that $\angle(Ax, x) = 0$ for all $x \in \mathbb{R}^n$, *i.e.*, x and Ax always point in the same direction. What can you say about the matrix A ? Be very specific.

- 5. Right inverses.** This problem concerns the specific matrix

$$A = \begin{bmatrix} -1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

This matrix is full rank (*i.e.*, its rank is 3), so there exists at least one right inverse. In fact, there are many right inverses of A , which opens the possibility that we can seek right inverses that in addition have other properties. For each of the cases below, either find a specific matrix $B \in \mathbb{R}^{5 \times 3}$ that satisfies $AB = I$ and the given property, or explain why there is no such B . In cases where there is a right inverse B with the required property, you must briefly explain how you found your B . You must also attach a printout of your Julia script or Jupyter notebook showing the verification that $AB = I$. (We'll be very angry if we have to type in your 5×3 matrix into Julia to check it.) When there is no right inverse with the given property, briefly explain why there is no such B .

- a) The second row of B is zero.
- b) The nullspace of B has dimension one.

- c) The third column of B is zero.
- d) The second and third rows of B are the same.
- e) B is upper triangular, *i.e.*, $B_{ij} = 0$ for $i > j$.
- f) B is lower triangular, *i.e.*, $B_{ij} = 0$ for $i < j$.

6. Some true/false questions. Determine if the following statements are true or false. What we mean by “true” is that the statement is true for all values of the matrices and vectors given. (You can assume the entries of the matrices and vectors are all real.) You can’t assume anything about the dimensions of the matrices (unless it’s explicitly stated), but you can assume that the dimensions are such that all expressions make sense. For example, the statement “ $A + B = B + A$ ” is true, because no matter what the dimensions of A and B (which must, however, be the same), and no matter what values A and B have, the statement holds. As another example, the statement $A^2 = A$ is false, because there are (square) matrices for which this doesn’t hold. (There are also matrices for which it does hold, *e.g.*, an identity matrix. But that doesn’t make the statement true.)

- a) If all coefficients (*i.e.*, entries) of the matrices A and B are nonnegative, and both A and B are onto, then $A + B$ is onto.
- b) $\text{null} \left(\begin{bmatrix} A \\ A + B \\ A + B + C \end{bmatrix} \right) = \text{null}(A) \cap \text{null}(B) \cap \text{null}(C).$
- c) $\text{null} \left(\begin{bmatrix} A \\ AB \\ ABC \end{bmatrix} \right) = \text{null}(A) \cap \text{null}(B) \cap \text{null}(C).$
- d) $\text{null}(B^T A^T A B + B^T B) = \text{null}(B).$
- e) If $\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$ is full rank, then so are the matrices A and B .
- f) If $\begin{bmatrix} A & 0 \end{bmatrix}$ is onto, then A is full rank.
- g) If A^2 is onto, then A is onto.
- h) If $A^T A$ is onto, then A is onto.
- i) Suppose $u_1, \dots, u_k \in \mathbb{R}^n$ are nonzero vectors such that $u_i^T u_j \geq 0$ for all i, j . Then the vectors are *nonnegative independent*, which means if $\alpha_1, \dots, \alpha_k \in \mathbb{R}$ are nonnegative scalars, and $\sum_{i=1}^k \alpha_i u_i = 0$, then $\alpha_i = 0$ for $i = 1, \dots, k$.
- j) Suppose $A \in \mathbb{R}^{n \times k}$ and $B \in \mathbb{R}^{n \times m}$ are skinny, full rank matrices that satisfy $A^T B = 0$. Then $\begin{bmatrix} A & B \end{bmatrix}$ is skinny and full rank.

7. Transmit powers in a wireless network. We consider a network of n transmitter/receiver pairs. Transmitter i transmits at power level p_i , which must satisfy $0 \leq p_i \leq P^{\max}$, where $P^{\max}$ is a given maximum transmitter power (which is the same for all transmitters). The path gain from transmitter j to receiver i is G_{ij} (which are all nonnegative, and G_{ii} are positive). The signal power at receiver i is given by $s_i = G_{ii}p_i$. The noise plus interference power at receiver i is given by

$$q_i = \sigma + \sum_{j \neq i} G_{ij}p_j$$

where $\sigma > 0$ is the self-noise power of the receivers (assumed to be the same for all receivers). The *signal to interference plus noise ratio* (SINR) at receiver i is defined as $S_i = s_i/q_i$.

- Explain how to determine if there is a power allocation (*i.e.*, a vector p) that satisfies the constraints $0 \leq p_i \leq P^{\max}$ and achieves $S_i = S^{\text{target}}$ for $i = 1, \dots, n$, where S^{target} is a (positive) target value of SINR. Explain how to find such a power allocation when it exists. You can assume that a matrix appearing in your analysis is full rank, but please make this assumption explicit.
- Among the SINR target values $S^{\text{target}} = 2, 2.1, 2.2, \dots, 3.9, 4$, find the largest for which there is a power allocation that satisfies the constraints $0 \leq p_i \leq P^{\max}$ and achieves $S_i = S^{\text{target}}$ for $i = 1, \dots, n$, for the problem data

$$G = \begin{bmatrix} 1 & .2 & .1 \\ .1 & 2 & .1 \\ .3 & .1 & 3 \end{bmatrix}, \quad \sigma = 0.01, \quad P^{\max} = 0.1.$$

Remarks.

- Yes, this problem includes constraints on p (*i.e.*, that its entries are nonnegative and no more than $P^{\max}$), which we have not covered in EE263. Still, you can solve it (with material we *have* covered).
- If you solve this problem using methods that are more advanced or complicated than needed, we will deduct points.

8. Interpolation with rational functions.. Consider a function $f : \mathbb{R} \rightarrow \mathbb{R}$ of the form

$$f(x) = \frac{a_0 + a_1x + \dots + a_mx^m}{1 + b_1x + \dots + b_mx^m},$$

where $a_0, \dots, a_m$ and $b_1, \dots, b_m$ are parameters, with either $a_m \neq 0$ or $b_m \neq 0$. Such a function is called a rational function of degree m . We are given data points $x_1, \dots, x_N \in \mathbb{R}$, and $y_1, \dots, y_N \in \mathbb{R}$, where $y_i = f(x_i)$.

- Explain how to find a rational function of smallest degree that is consistent with the data: that is, explain how to find the smallest value of m , and corresponding values of $a_0, \dots, a_m$, and $b_1, \dots, b_m$ such that $f(x_i) = y_i$ for $i = 1, \dots, N$.
- Carry out your method on the data in `rational_interpolation_data.json`. Report your value of m , and the corresponding coefficients $a_0, \dots, a_m$, and $b_1, \dots, b_m$. Plot the data and the rational function $f(x)$. Verify that $y_i = f(x_i)$ for $i = 1, \dots, N$ (possibly with small numerical errors).

9. (Optional) Price elasticity of demand. The demand for n different goods is a function of their prices:

$$q = f(p),$$

where p is the price vector, q is the demand vector, and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the demand function. The current price and demand are denoted p^* and q^* , respectively. Now suppose there is a small price change δp , so $p = p^* + \delta p$. This induces a change in demand, to $q \approx q^* + \delta q$, where

$$\delta q \approx Df(p^*)\delta p,$$

where Df is the derivative or Jacobian of f , with entries

$$Df(p^*)_{ij} = \frac{\partial f_i}{\partial p_j}(p^*).$$

This is usually rewritten in term of the *elasticity matrix* E , with entries

$$E_{ij} = \frac{\partial f_i}{\partial p_j}(p^*) \frac{1/q_i^*}{1/p_j^*},$$

so E_{ij} gives the relative change in demand for good i per relative change in price j . Defining the vector y of relative demand changes, and the vector x of relative price changes,

$$y_i = \frac{\delta q_i}{q_i^*}, \quad x_j = \frac{\delta p_j}{p_j^*},$$

we have the linear model $y = Ex$.

Here are the questions:

- a) What is a reasonable assumption about the diagonal elements E_{ii} of the elasticity matrix?
- b) Goods i and j are called *substitutes* if they provide a similar service or other satisfaction (*e.g.*, train tickets and bus tickets, cake and pie, etc.). They are called *complements* if they tend to be used together (*e.g.*, automobiles and gasoline, left and right shoes, etc.). For each of these two generic situations, what can you say about E_{ij} and E_{ji} ?
- c) Suppose the price elasticity of demand matrix for two goods is

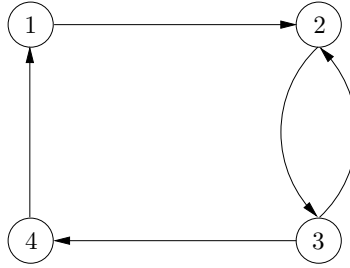
$$E = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}.$$

Describe the nullspace of E , and give an interpretation (in one or two sentences). What kind of goods could have such an elasticity matrix?

10. (Optional) Paths and cycles in a directed graph. We consider a directed graph with n nodes. The graph is specified by its *node adjacency matrix* $A \in \mathbb{R}^{n \times n}$, defined as

$$A_{ij} = \begin{cases} 1 & \text{if there is an edge from node } j \text{ to node } i \\ 0 & \text{otherwise.} \end{cases}$$

Note that the edges are *oriented*, i.e., $A_{34} = 1$ means there is an edge from node 4 to node 3. For simplicity we do not allow self-loops, i.e., $A_{ii} = 0$ for all i , $1 \leq i \leq n$. A simple example illustrating this notation is shown below.



The node adjacency matrix for this example is

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

In this example, nodes 2 and 3 are connected in both directions, i.e., there is an edge from 2 to 3 and also an edge from 3 to 2. A *path* of length $l > 0$ from node j to node i is a sequence $s_0 = j, s_1, \dots, s_l = i$ of nodes, with $A_{s_{k+1}, s_k} = 1$ for $k = 0, 1, \dots, l-1$. For example, in the graph shown above, 1, 2, 3, 2 is a path of length 3. A *cycle* of length l is a path of length l , with the same starting and ending node, with no repeated nodes other than the endpoints. In other words, a cycle is a sequence of nodes of the form $s_0, s_1, \dots, s_{l-1}, s_0$, with

$$A_{s_1, s_0} = 1, \quad A_{s_2, s_1} = 1, \quad \dots \quad A_{s_0, s_{l-1}} = 1,$$

and

$$s_i \neq s_j \text{ for } i \neq j, \quad i, j = 0, \dots, l-1.$$

For example, in the graph shown above, 1, 2, 3, 4, 1 is a cycle of length 4. The rest of this problem concerns a specific graph, given in the file **directed_graph.json** on the course web site. For each of the following questions, you must give the answer explicitly (for example, enclosed in a box). You must also explain clearly how you arrived at your answer.

- What is the length of a shortest cycle? (Shortest means minimum length.)
- What is the length of a shortest path from node 13 to node 17? (If there are no paths from node 13 to node 17, you can give the answer as ‘infinity’.)
- What is the length of a shortest path from node 13 to node 17, that *does not* pass through node 3?
- What is the length of a shortest path from node 13 to node 17, that *does* pass through node 9?
- Among all paths of length 10 that start at node 5, find the most common ending node.
- Among all paths of length 10 that end at node 8, find the most common starting node.
- Among all paths of length 10, find the most common pair of starting and ending nodes. In other words, find q, r which maximize the number of paths of length 10 from q to r .